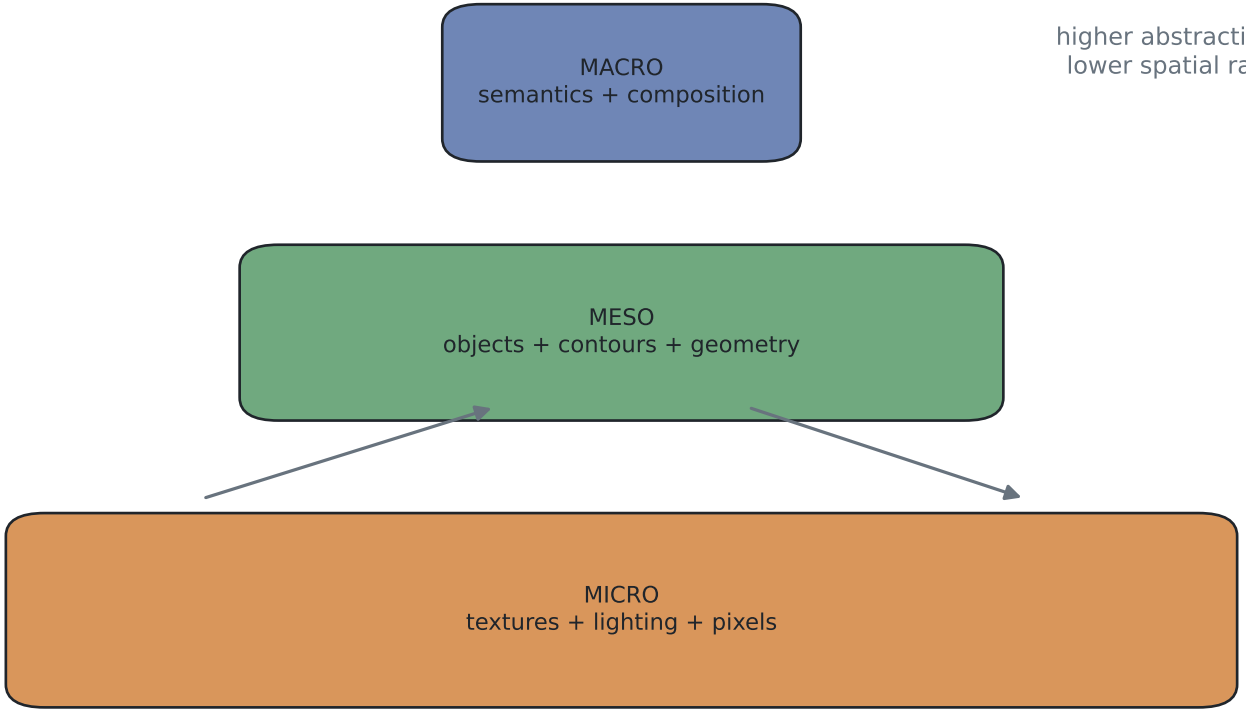Hierarchical decomposition of visual state